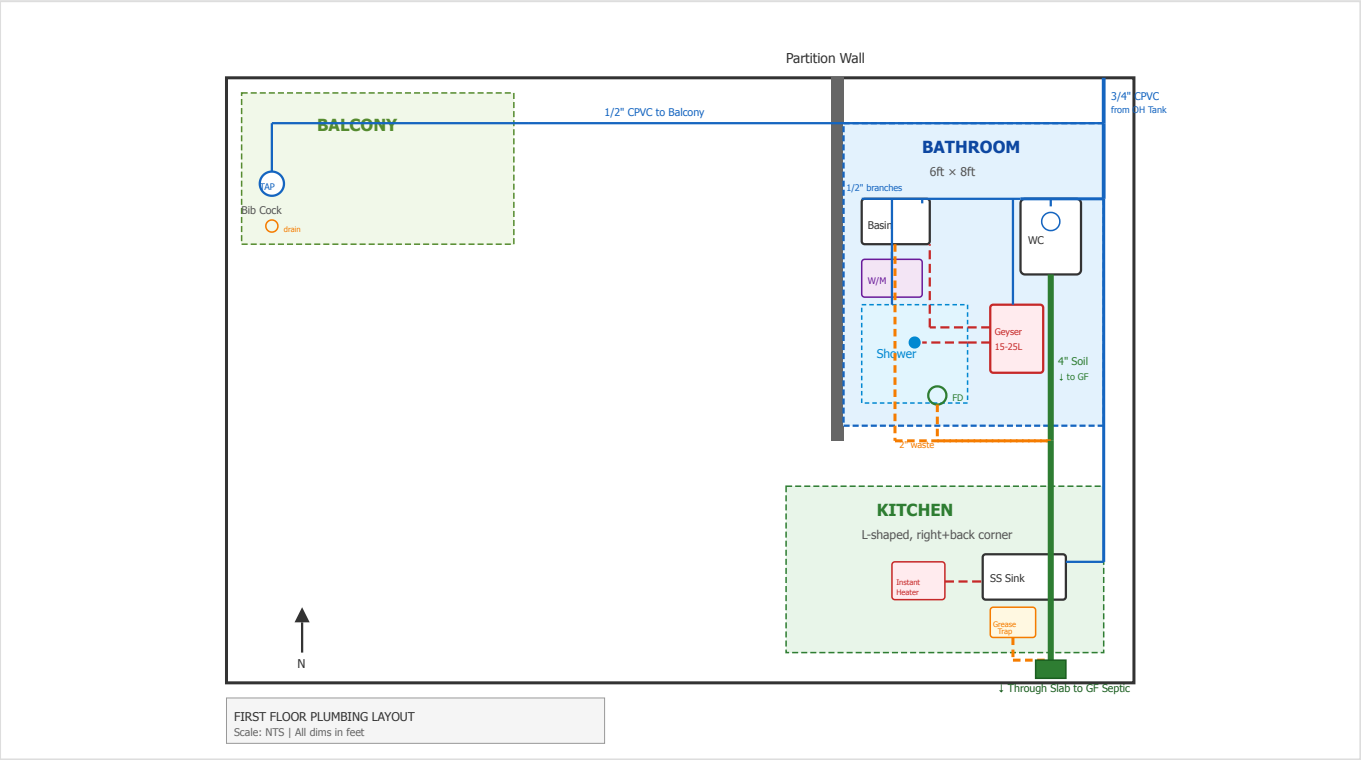


पहली मंजिल — Plumbing Design

First Floor Plumbing Layout | प्लंबिंग डिज़ाइन | Print-Ready Document

1. Plumbing Layout Plan / प्लंबिंग लेआउट

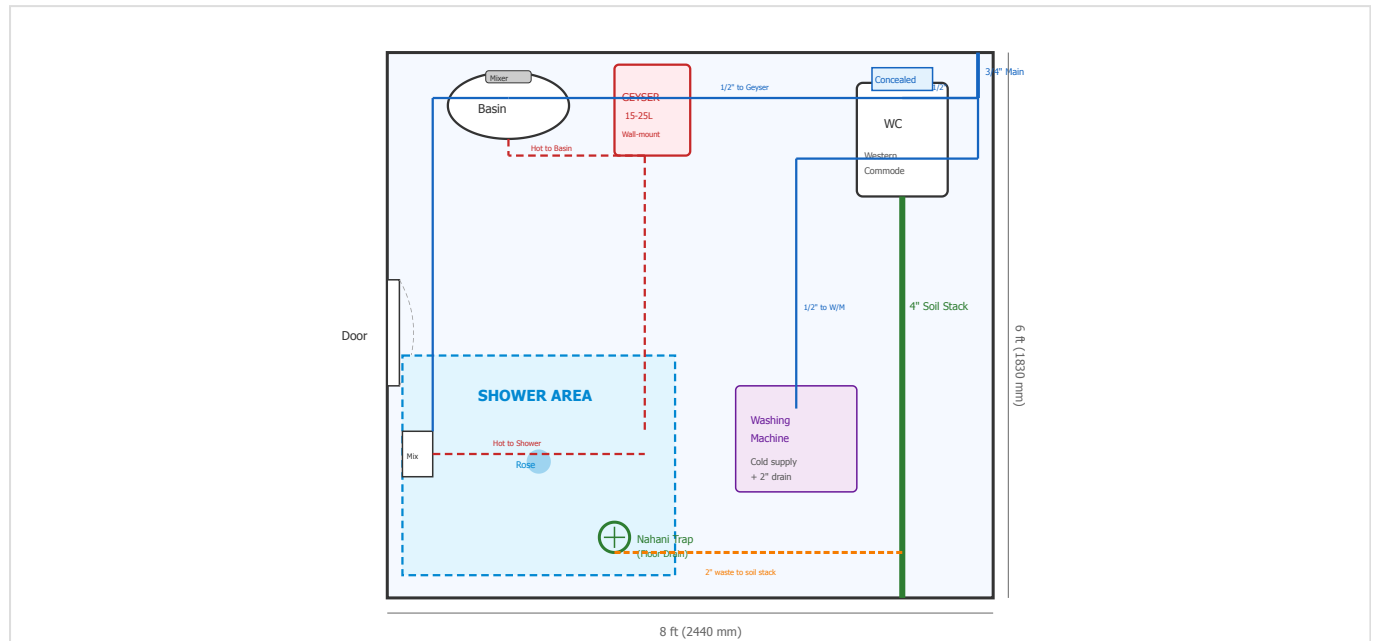
- Cold Water Supply (CPVC)
- Hot Water Supply (CPVC)
- Soil Pipe 4" (PVC SWR)
- Waste Pipe 2" (PVC SWR)
- Vent Pipe



Note / नोट: All supply pipes run concealed in walls (chase cutting). Drainage pipes run below slab soffit (exposed or in false ceiling). Main 4" soil stack is on RIGHT wall — verify sleeve position in slab.

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2. Bathroom Detail / बाथरूम विवरण (6ft × 8ft)



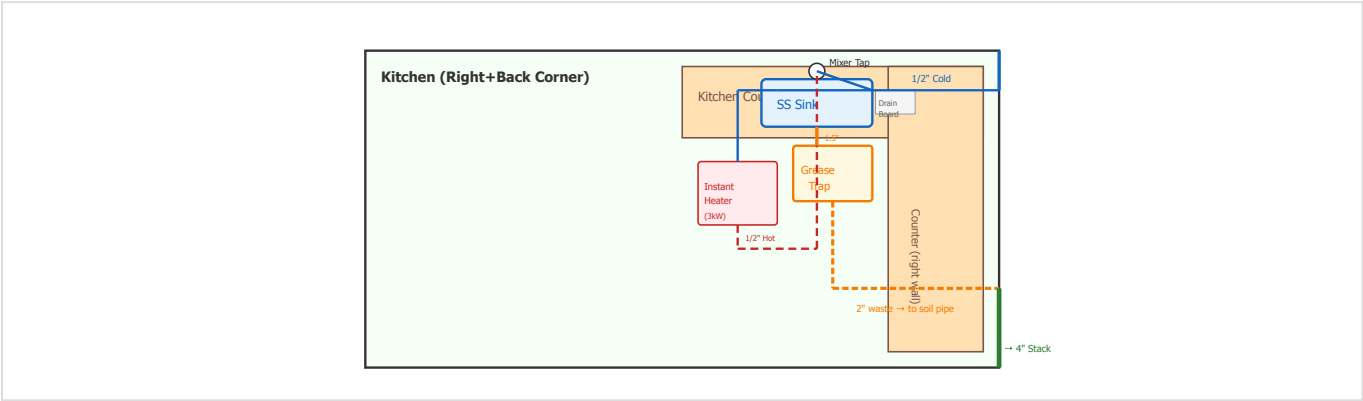
Bathroom Fixture Heights (from FFL / फ्लोर से ऊंचाई):

Fixture / फिक्स्चर	Supply Point Height	Drain Point Height	Notes
WC (Concealed Flush)	1000mm (cold inlet at back)	Floor level (150mm center)	Flush tank concealed in wall at 800-1100mm
Wash Basin	600mm (H+C mixer)	500mm (32mm waste)	Wall-hung basin at 800mm from floor
Shower Mixer	1100mm (H+C)	Floor level (nahani)	Overhead rose at 2000mm, hand shower at 1000mm
Geyser (15-25L)	1800mm (cold in, hot out)	—	Wall-mounted, needs 6mm ² power point at 2000mm
Washing Machine	800mm (cold only, 15mm)	Floor level (floor trap)	Angle valve with hose connection
Nahani Trap (Floor Drain)	—	Floor level	Jali type SS, 110mm, connects to 2" waste

Hot Water Routing / गर्म पानी: Geyser → Basin (short run ~1m) + Shower (~2m). Insulate hot pipes with foam sleeve to reduce heat loss. Geyser placement on wall between basin and shower minimizes pipe length.

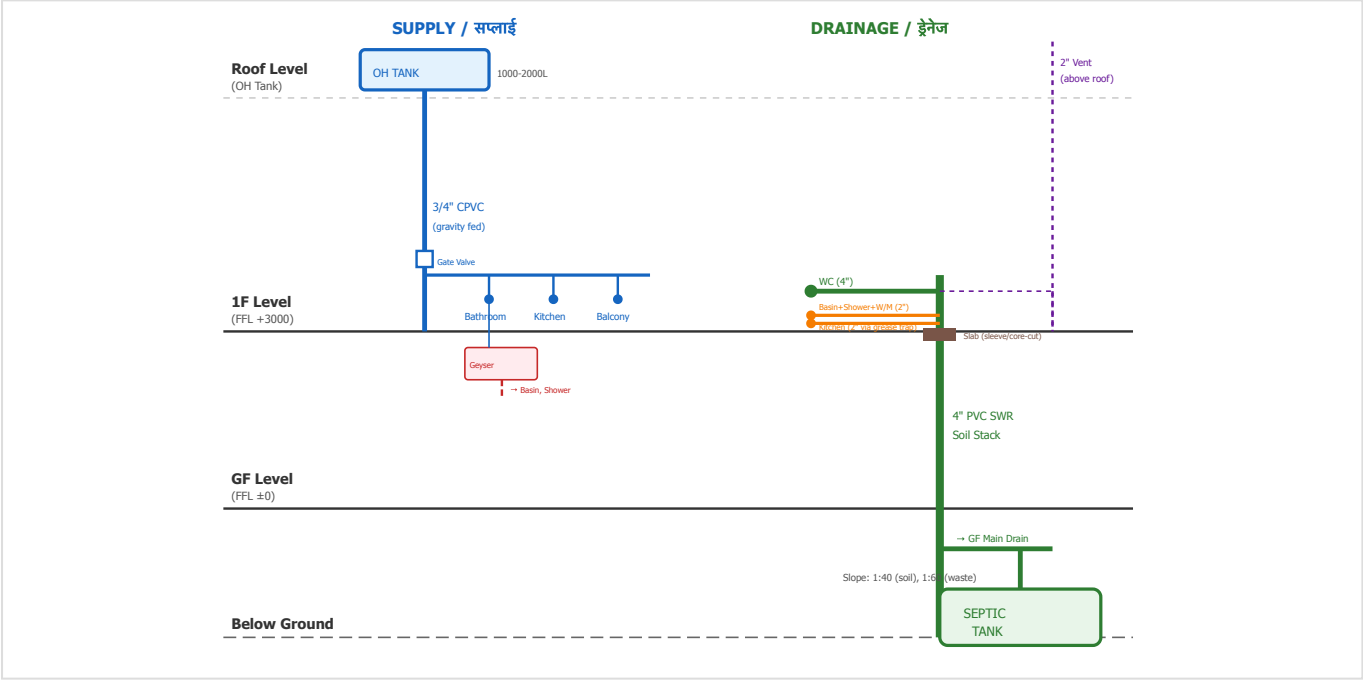
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3. Kitchen Plumbing Detail / किचन प्लंबिंग



Grease Trap / ग्रीस ट्रैप: Install under-counter grease trap (capacity 10-15L) between sink waste and main drain pipe. Clean every 2 weeks. This prevents kitchen oil/grease from entering main drainage and clogging pipes.

4. Riser Diagram / राइज़र डायग्राम (Vertical Section)



CRITICAL / ज़रूरी: Check if pipe sleeve (4" dia) was left in slab during GF ceiling pour on the RIGHT side. If not present, core-cutting (4" hole) will be needed before 1F wall construction begins. Plan this FIRST.

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5. Pipe Sizing Table / पाइप साइज टेबल

S.No.	Pipe / पाइप	Size	Material	Slope	Purpose / उपयोग	Location
1	Main Supply (OH Tank → 1F)	3/4" (20mm)	CPVC (SDR 11)	—	Main cold water supply from overhead tank	Vertical riser, right wall
2	Branch Supply — Bathroom	1/2" (15mm)	CPVC (SDR 11)	—	Cold water to WC, basin, shower, geyser, W/M	Concealed in bathroom walls
3	Branch Supply — Kitchen	1/2" (15mm)	CPVC (SDR 11)	—	Cold water to kitchen sink + instant heater	Concealed in kitchen wall
4	Branch Supply — Balcony	1/2" (15mm)	CPVC (SDR 11)	—	Cold water to balcony bib cock	Concealed, front/left wall
5	Hot Water (Geyser → Fixtures)	1/2" (15mm)	CPVC (SDR 11)	—	Hot water to basin + shower mixer	Concealed, bathroom walls, insulated
6	Hot Water (Instant Heater → Kitchen)	1/2" (15mm)	CPVC (SDR 11)	—	Hot water to kitchen sink mixer	Concealed, kitchen wall
7	Soil Pipe (WC)	4" (110mm)	PVC SWR (Type B)	1:40	WC waste + combined drain to GF	Vertical stack, right side, through slab
8	Waste Pipe — Basin	1.25" (32mm)	PVC SWR	1:50	Basin waste to floor trap / soil stack	Concealed / below slab soffit
9	Waste Pipe — Shower/Nahani	2" (50mm)	PVC SWR	1:60	Floor drain waste to soil stack	Below slab / in screed
10	Waste Pipe — Kitchen Sink	2" (50mm)	PVC SWR	1:60	Kitchen waste (post grease trap) to soil stack	Below slab / along wall
11	Waste Pipe — Washing Machine	2" (50mm)	PVC SWR	1:60	W/M drain to floor trap	Floor level in bathroom
12	Vent Pipe	2" (50mm)	PVC SWR	—	Anti-siphon vent, extends above roof	Connected to soil stack, rises through roof
13	Balcony Drain	2" (50mm)	PVC SWR	1:60	Balcony tap drain / rainwater	Through parapet, external downpipe

6. Fixture Schedule / फिक्स्चर शेड्यूल

S.No.	Fixture / फिक्स्चर	Type / Brand Category	Supply Connection	Drain Connection	Location
1	Western Commode (WC)	Floor-mounted, S-trap, concealed cistern (Hindware/Parryware/Jaquar)	15mm cold (concealed valve)	4" (110mm) soil	Bathroom — right wall
2	Concealed Flush Tank	Slim cistern, dual-flush (3/6L) (Grohe Rapid / Geberit type)	15mm cold (from above)	Connected to WC	Inside partition wall behind WC
3	Wash Basin	Wall-hung ceramic, table-top or wall-mount (Hindware/Cera/Parryware)	15mm H + 15mm C (mixer)	32mm waste with bottle trap	Bathroom — left wall
4	Basin Mixer Tap	Single-lever basin mixer (Jaquar/Grohe)	15mm H + C (from wall)	—	Above basin
5	Shower Mixer	Concealed diverter mixer (hot+cold)	15mm H + C (concealed)	—	Shower area wall

		with overhead rose + hand shower			
6	Nahani Trap (Floor Drain)	SS jali, cockroach trap 110mm square/round	—	50mm (2") waste	Shower area floor (lowest point)
7	Storage Geyser	15-25L, wall-mounted, vertical (Racold/AO Smith/Havells)	15mm cold in, 15mm hot out	Safety valve drain to floor	Bathroom wall (high, 1800mm)
8	Kitchen Sink	SS single bowl + drainboard 24"×18" (600×450mm)	15mm H + C (mixer)	40mm waste → grease trap	Kitchen counter — back wall
9	Kitchen Mixer Tap	Single-lever sink mixer, swivel spout (Jaquar/Grohe/Hindware)	15mm H + C	—	Kitchen wall above sink
10	Instant Water Heater (Kitchen)	3kW, wall-mounted (Racold/Bajaj)	15mm cold in, 15mm hot out	—	Kitchen wall near sink
11	Grease Trap	Under-counter, 10-15L capacity SS or PVC type	—	40mm in, 50mm out	Below kitchen sink
12	Washing Machine Point	Angle valve + multi-turn cock with hose connector	15mm cold (angle valve)	50mm floor trap	Bathroom / adjacent area
13	Balcony Bib Cock	Brass bib cock (15mm) (Jaquar/Parryware)	15mm cold	Small floor drain below	Left/front wall of balcony

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7. Material List / सामान की सूची (Estimated Quantities)

A. Supply Pipes (CPVC) / सप्लाई पाइप

Item	Size	Qty (approx)	Remarks
CPVC Pipe (Astral/Supreme)	3/4" (20mm)	6m (2 lengths)	Main supply from OH tank to distribution
CPVC Pipe (Astral/Supreme)	1/2" (15mm)	30m (10 lengths)	All branch supply lines (cold + hot)
CPVC Elbows 90°	1/2"	25 nos	Direction changes
CPVC Elbows 90°	3/4"	6 nos	Main line turns
CPVC Tees	1/2"	12 nos	Branch take-offs
CPVC Tees (Reducing)	3/4" × 1/2"	4 nos	Main to branch reduction
CPVC Couplers	1/2"	10 nos	Pipe joints
CPVC MTA (Male Threaded Adapter)	1/2"	14 nos	Fixture connections
CPVC FTA (Female Threaded Adapter)	1/2"	8 nos	Valve connections
Gate Valve (brass)	3/4"	1 no	Main supply shut-off
Concealed Stop Valves	1/2"	8 nos	At each fixture point (for isolation)
Angle Valves	1/2"	4 nos	WC, W/M, geyser connections
CPVC Solvent Cement	—	2 tins (100ml)	One-step cement for CPVC joints
Pipe Insulation Foam	1/2" ID	6m	Hot water pipe insulation
Pipe Clamps / Clips	1/2" + 3/4"	30 nos	Wall fixing before plastering

B. Drainage Pipes (PVC SWR) / ड्रेनेज पाइप

Item	Size	Qty (approx)	Remarks
PVC SWR Pipe (Prince/Supreme)	4" (110mm)	6m (2 lengths)	Soil stack (vertical drop through slab)
PVC SWR Pipe	2" (50mm)	12m (4 lengths)	Waste pipes — shower, kitchen, W/M, balcony
PVC SWR Pipe	1.25" (32mm)	2m	Basin waste (short run to trap/stack)
PVC SWR Elbows 90°	4"	3 nos	Soil pipe bends
PVC SWR Elbows 45°	4"	2 nos	Gradual direction changes (preferred)
PVC SWR Elbows 90°	2"	8 nos	Waste pipe turns
PVC SWR Tees / Y-branches	4" × 2"	3 nos	Waste pipe connection to soil stack
PVC SWR Tees	2"	3 nos	Waste pipe junctions
Nahani Trap (Floor Trap) with jali	110mm	2 nos	Bathroom floor + W/M drain
P-Trap / Bottle Trap (Basin)	32mm	1 no	Wash basin waste trap
WC Connector (offset if needed)	4" (110mm)	1 no	WC to soil pipe connection
Cleaning Eye / Access Plug	4"	2 nos	For maintenance access on soil stack
PVC Solvent Cement	—	2 tins (200ml)	SWR pipe joints
Vent Cowl	2"	1 no	Top of vent pipe (prevents debris entry)
Pipe Sleeve (MS)	4" + 2"	2+2 nos	Through-slab sleeves (if not already cast)

C. Fixtures & Accessories / फिक्स्चर

Item	Qty	Remarks
Western Commode (S-trap, floor-mount)	1	With seat cover, fixing kit
Concealed Flush Tank (dual-flush)	1	With flush plate (SS or chrome)
Wash Basin (wall-hung/table-top)	1	With fixing brackets
Basin Mixer Tap (single-lever)	1	Hot + cold
Shower Mixer (concealed diverter)	1	With overhead rose + hand shower + pipe
Storage Geyser (15-25L)	1	With safety valve, mounting bracket
SS Kitchen Sink (single bowl + drainboard)	1	With waste coupling + basket strainer
Kitchen Sink Mixer (swivel)	1	Hot + cold, wall-mount or deck-mount
Instant Water Heater (3kW)	1	Kitchen — wall mounted
Grease Trap (10-15L)	1	Under-counter, SS or PVC
Bib Cock (brass, 15mm)	1	Balcony tap
Washing Machine Valve (angle)	1	With hose connector
Teflon Tape (thread seal)	5 rolls	All threaded connections

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8. Contractor Notes / ठेकेदार के लिए निर्देश

Phase 1: Before Wall Construction / दीवार बनाने से पहले

- **Slab Sleeve Check (स्लैब स्लीव जांच):** Verify if 4" pipe sleeve exists on RIGHT side of slab. If not, mark and core-cut a 4" (110mm) hole BEFORE starting walls. Also need 2" sleeve for kitchen waste if separate.
- **Soil Stack Position (सॉइल पाइप स्थान):** Mark exact position of 4" vertical soil pipe on the slab — should be against right wall, close to where WC will be.
- **Floor Trap Positions (फ्लोर ट्रैप):** Set nahani traps in concrete/screed BEFORE floor tiling. Bathroom floor must slope toward drain (1:100 minimum).
- **Drainage Pipe Below Slab:** Run horizontal waste pipes (2") from kitchen and bathroom to the soil stack position BEFORE screeding the floor (if pipes run in floor). Otherwise, run them below the slab soffit (exposed/in false ceiling at GF).

Phase 2: During Wall Construction / दीवार बनाते समय

- **Pipe Chases (पाइप की नाली):** Leave vertical chases in walls for concealed supply pipes — 50mm wide × 40mm deep for 1/2" pipe; 60mm wide × 50mm deep for 3/4" pipe. Do NOT chase after building — weakens wall.
- **Concealed Cistern Space:** Build partition wall behind WC at minimum 120mm thickness (or use double-wall technique) to house concealed flush tank. Frame the mounting plate at correct height (800-1000mm from FFL).
- **Geyser Wall:** Use solid block/brick behind geyser location. Need 4 nos heavy-duty wall plugs at 1800mm height. Also run concealed 6mm² power cable to a 15A switch at 2000mm.
- **Kitchen Wall:** Leave chase for hot + cold supply pipes behind sink location.

Phase 3: After Walls, Before Plastering / प्लास्टर से पहले

- **Pipe Installation (पाइप डालना):** Fix all CPVC supply pipes in wall chases. Test entire system at 2× working pressure (6 kg/cm² for min 2 hours) before covering.
- **Pressure Test (प्रेसर टेस्ट):** MANDATORY — fill system with water, pressurize to 6 bar, hold for 2 hours. No drop = pass. Mark any leaks, fix, and re-test.
- **Photo Documentation:** Take photos of ALL concealed pipe routes before plastering — will be needed later for any maintenance.
- **Clip Pipes:** Secure all pipes with wall clamps every 600mm (horizontal) and 1000mm (vertical) before plastering over.
- **Cover Chases:** Fill chases with chicken mesh + cement mortar after pipe testing passes.

Phase 4: After Plastering, Before Tiling / टाइल्स से पहले

- **Floor Slopes:** Bathroom floor must slope toward nahani trap — minimum 12mm fall per meter (1:80). Verify with water test before tiling.
- **Waterproofing:** Complete bathroom waterproofing (floor + walls 1ft above floor) AFTER drainage pipe installation and BEFORE tiling.
- **Fixture Points:** Mark and verify all fixture outlet points match the chosen fixtures (height, spacing between H+C points = 150mm standard for mixers).

Phase 5: After Tiling / टाइल्स के बाद

- **Fixture Installation:** Install WC, basin, shower fittings, kitchen sink, balcony tap.
- **Seal All Joints:** Silicone sealant around WC base, basin wall junction, and kitchen sink edges.
- **Final Testing:** Run all fixtures simultaneously — check flow, drainage speed, any leaks. Run for 30 minutes minimum.
- **Grease Trap:** Install under kitchen counter after sink is fixed. Ensure easy access for cleaning.

IMPORTANT CHECKS / महत्वपूर्ण जांच:

- All CPVC joints must use ONE-STEP solvent cement (not two-step primer). Allow 24hr cure before pressure test.

- PVC SWR joints: use SWR solvent cement (different from CPVC). Check ring-seal rubber is in place for push-fit joints.
- Drainage slope: soil pipe 1:40 minimum, waste pipe 1:60 minimum. USE SPIRIT LEVEL.
- Vent pipe MUST extend minimum 150mm above roof level with cowl. Do not terminate inside building.
- Keep cleaning eyes (access plugs) accessible — do not plaster/tile over them.

Coordination / तालमेल:

- Plumber + Mason: Chase cutting must be done during wall construction, NOT after.
- Plumber + Electrician: Geyser power point (15A) and instant heater point (20A) at correct locations BEFORE plastering.
- Plumber + Waterproofing: Complete drain installations BEFORE waterproofing. Do NOT puncture waterproofing membrane after application.
- Plumber + Tiler: Coordinate fixture outlet heights with tile layout (cuts, patterns).

Document Info: First Floor Plumbing Design | Project: Farmhouse | Prepared for construction reference

Standards Reference: IS 2065 (CPVC pipes), IS 13592 (PVC SWR), IS 2556 (Sanitary fixtures), NBC 2016 Part 9

Note: All quantities are estimates — final measurement to be done on-site. Allow 10-15% extra for wastage/cuts.